

UDOXAN INJECTION (1ML AMP)

DESCRIPTION: UDOXAN INJECTION is a clear, colourless solution.

COMPOSITION: Each ampoule contains Oxytocin (synthetic) 10 I.U. per ml.

PHARMACODYNAMICS: Oxytocin stimulates the smooth muscle of the uterus, producing rhythmic contractions. It also causes contraction of the myo-epithelial cells surrounding the mammary alveoli. It therefore facilitates lactation in women experiencing difficulty in breast feeding. Synthetic oxytocin has only very slight pressor and antidiuretic activity (absence of vasopressin). By increasing calcium concentration, oxytocin stimulates uterine smooth muscle contractions and thus behaves like contraction of normal, spontaneous labour and transiently impeding uterine blood flow. Amplitude and duration of uterine contractions are increased leading to dilatation and effacement of the cervix. Throughout pregnancy there is a gradual increase of oxytocin receptors and therefore uterine response, reaching its peak at term. Oxytocin stimulates smooth muscle to facilitate ejection of milk from breasts.

PHARMACOKINETICS:

Oxytocin, being a polypeptide, is largely inactivated in the alimentary tract and therefore virtually ineffective when ingested. It is however, effectively absorbed by the nasal mucosa and acts via the blood stream.

- Low protein binding (30%)
- Biotransformation: Primarily by tissue oxytocinase via enzymatic hydrolysis. Oxytocinase is also found in placental tissue and plasma.
- Half-life: 1 to 6 minutes (decreased in late pregnancy and lactation).
- Onset of action: Intramuscular - 3 to 5 minutes
Intravenous - Immediate
- Duration of action: Intramuscular - 2 to 3 hours
Intravenous - Uterine activity generally subsides within one hour.
- Elimination: Only small amounts are excreted unchanged.

INDICATION: Induction of labour; inadequate uterine effort; management of third stage of labour; post-partum haemorrhage.

RECOMMENDED DOSAGE:

Ampoules: For the induction and enhancement of labour, oxytocin should be given as an intravenous infusion and not by subcutaneous, intramuscular or intravenous bolus injection.

Infusion Fluids: Compatibility of oxytocin has been demonstrated with 0.9% saline and 5% dextrose solution. Oxytocin is not compatible with solutions containing bisulphites and metabisulphites as preservatives. Due attention should be paid to the choice of infusion fluid in individual patients. Generally, oxytocin should be administered in a combination of dextrose and an electrolyte solution (such as 4 % in N/5 saline), or in an isotonic electrolyte solution. The use of 5% dextrose in water is not recommended.

Induction or enhancement of labour: Oxytocin should be administered as an intravenous infusion, preferably by means of variable speed infusion pump, or by drip infusion.

The initial infusion rate should be set at 1 to 4 mU/min. This rate may be gradually increased at intervals of not shorter than 20 min. until a contraction pattern similar to that of normal labour is established (max. rate: 20 mU/min). The increments in infusion rate should not be as high once contractions have been established as those used to initiate contractions. The frequency and duration of contractions and foetal heart rate must be carefully monitored during oxytocin administration, the latter preferably by electronic means, and the infusion must be discontinued immediately in the event of uterine hyperactivity or foetal distress or foetal heart abnormalities.

If regular contractions are not established after the infusion of 5 I.U. oxytocin, the attempt to induce labour should be broken off, it can generally be repeated on the following day.

For drip infusion, the preparation of a solution containing 10 I.U. oxytocin per 1 Litre infusion fluid is recommended. To ensure even mixing of the drip solution, the bottle or bag must be turned upside down several times before use. Using this concentration, the recommended initial infusion rate of 1-4 mU/min corresponds to 0.1-0.4 mL/min (= 2-8 drops/min), and the recommended maximum rate of 20 mU/min is reached at a rate of 2 mL/min (= 40 drops/min).

When using a mechanical infusion pump which delivers smaller volume than those given by drip infusion, a more concentrated oxytocin solution will be required. The concentration suitable for infusions within the recommended dosage range (1-20 mU/min) must be calculated according to the specification of the pump used.

Augmentation of labour: In general, the dose of Oxytocin required for the augmentation of labour is less than that required for induction. Therefore, the initial infusion rate should be at the lower end of the recommended range.

Third stage of labour and puerperium (haemorrhage, subinvolution of the uterus): 5-10 I.U. i.m. or 5 I.U. i.v. In patients given oxytocin by drip to induce or stimulate labour, the infusion should be continued during the third stage.

Caesarian section: 5 I.U. intramuscular after delivery of the foetus.

ROUTE OF ADMINISTRATION: Sterile solution for i.v. infusion or slow i.v. and i.m injection

CONTRAINDICATIONS:

Oxytocin is contraindicated in any of the following conditions:

Significant cephalopelvic disproportion; unfavourable fetal positions or presentations which are undeliverable without conversion prior to delivery, eg. transverse lies; in obstetrical emergencies where the benefit-to-risk ratio for either the fetus or the mother favors surgical intervention; in cases of fetal distress where delivery is not imminent; hypertonic uterine patterns; hypersensitivity to the drug.

Prolonged use in uterine inertia or severe toxemia is contraindicated.

Oxytocin should not be used cases where vaginal delivery is not indicated, such as cord presentation or prolapse, total placenta previa, and vasa previa.

Excessive distension of the uterus (e.g. multiple pregnancy, hydramnios), parity greater than 4, elderly multiparae, previous Caesarean section or other surgery involving the uterus.

Predisposition to amniotic fluid embolism (foetal death-in-utero, abruptio placentae), hypertonic contractions,

WARNINGS & PRECAUTIONS:

Use in pregnancy: Use of this drug has contributed significantly to the safety of parturition. However, there have been instances of idiosyncratic sensitivity of the uterus resulted in foetal anoxia.

The induction of labour by means of oxytocic agents should be attempted only when strictly indicated for medical reasons rather than for convenience. Administration should only be under hospital conditions, and all patients receiving intravenous oxytocin must be under continuous observation by trained personnel with a thorough knowledge of the drug and qualified to identify complications. A physician qualified to manage any complications must be immediately available.

When oxytocin is given for induction and enhancement of labour, it must only be administered as an I.V. infusion preferably by means of a motor-driven variable speed infusion pump, and not by subcutaneous, intramuscular or intravenous bolus injection. Careful monitoring is essential (foetal heart rate, uterine response – by tocometry if possible, blood pressure), so that the dosage may be titrated to individual response.

When oxytocin is used for the induction of labour, there is a chance that infants with unanticipated prematurity may be delivered. To reduce this risk, it is recommended that in women with uncertain obstetric dating, the maturity of the foetus be assessed by ultrasonic measurement of foetal biparietal diameter.

When oxytocin is used for the management of the third stage of labour, multiple pregnancy must be excluded before the drug is injected.

Oxytocin should not be given parenterally at the same time as oxytocin nasal spray.

In patients with cardiovascular disorders the infusion volume be kept low by using a more concentrated solution. In solutions where more prolonged periods of administration may be required, such as earlier in gestation when the uterus is less sensitive to oxytocin, special precautions must be taken to avoid water intoxication.

Water intoxication: Water intoxication, which is potentially fatal can result from high doses or prolonged periods of infusion of oxytocin in electrolyte-free fluids. The weak anti-diuretic activity of oxytocin acting to increase water reabsorption from the glomerular filtrate, may be a contributing factor, but the major cause is the use of large amounts of electro-free fluids.

The symptoms and signs of water intoxication are:

1. Headache, anorexia, nausea, vomiting and abdominal pain.
2. Lethargy, drowsiness, unconsciousness and grand mal type seizures.
3. Low blood electrolyte concentration.

Therefore, if high doses or a more prolonged period of administration is expected, the following precautions must be observed:

1. A strict fluid balance chart must be kept. Low-sodium infusion fluids should be avoided.
2. Oxytocin should be infused in small volumes of isotonic fluid, using higher concentrations that recommended for the induction and enhancement of labour in non-complicated cases.
3. Fluid intake by mouth must be restricted.
4. Maternal serum electrolytes should be measured at regular intervals e.g. 8 – 12 hourly.

Treatment of water intoxication:

1. Discontinuation oxytocin.
2. Restrict fluid intake.
3. Promote diuresis.
4. Correct electrolyte imbalance.
5. Control convulsions, e.g. with judicious use of diazepam.
6. If coma is present: maintain a free airway, and carry out the routine measures for care of an unconscious patient.

INTERACTION WITH OTHER MEDICAMENTS:

Prostaglandins may potentiate the effect of oxytocin and very careful monitoring is therefore recommended in cases of concomitant administration.

Some inhaled anaesthetics, eg. cyclopropane or halothane, may enhance the hypotensive effect of oxytocin and reduce its oxytocic action. Their concurrent use with oxytocin has also been reported to cause cardiac rhythm disturbances.

When given during or after caudal block anaesthesia, oxytocin may potentiate the pressor effect of sympathomimetic vasoconstrictor agents.

Concurrent use with other oxytocics in absence of adequate cervical dilatation may cause uterine hypertonus resulting in uterine rupture or cervical laceration. During concurrent use the patient should be monitored closely. As an adjunct to abortifacients, it is recommended to be used only after the effect of the abortifacient has subsided, in order to reduce the risk of uterine rupture and cervical laceration.

Water intoxication may also occur if oxytocin is given following the use of intra-amniotic hypertonic saline for abortion.

SIDE EFFECTS:

Cardiovascular: Hypotension, tachycardia and ECG changes have been observed following i.v. administration of concentrated solution. Rapid intravenous injection of as little as 2 I.U. can cause acute hypotension. Hypertension may occur rarely.

Other Reactions: Water intoxication (see **Precautions**) and associated neonatal hyponatraemia. Neonatal jaundice, neonatal convulsions, skin rashes and anaphylactoid reactions, headache, dyspnea, oedema, cardiovascular spasm and collapse have been reported occasionally.

Amniotic fluid embolism (in association with pre-disposing factors, e.g. tumultuous labour, foetal death-in-utero, meconium-stained amniotic fluid) have been reported.

Increases in post-partum bleeding have been observed rarely in conjunction with oxytocin. This effect probably relates more to abnormality of uterine action rather than to side effects of the drug.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage may give rise to the following complications:

- foetal distress (foetal bradycardia, meconium staining of the amniotic fluid, fetal asphyxia)
- uterine hypertonicity, tetanic contraction or rupture
- placental abruption
- amniotic fluid embolism
- water intoxication (see **Precautions**)

IV bolus injection or more than 2 I.U. can cause acute hypotension.

Treatment of overdosage:

1. The infusion should be immediately discontinued and the infusion line occluded at the point of entry into the skin.

2. Oxygen should be administered to the mother.

3. In the rare event that exaggerated contractions persist after stopping the infusion, a selective beta-2-adrenergic stimulant such as ritodrine can be used parenterally to achieve uterine relaxation, in order to relieve foetal distress.

The mother may be placed in the left lateral position.

Treatment of water intoxication: see **Precautions**.

INSTRUCTIONS FOR USE:Incompatibilities

Oxytocin should not be infused via the same apparatus as blood or plasma, because the peptide linkages are rapidly inactivated by oxytocin-inactivating enzymes. Oxytocin is incompatible with solutions containing sodium metabisulphite as a stabiliser.

Special precautions for disposal and other handling

Oxytocin is compatible with the following infusion fluids: sodium chloride 0.9 %, dextrose 5 %, Ringer's solution, acetated Ringer's solution.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

STORAGE CONDITIONS:

Store in a cool dry place between 2°C-8°C. Protect from light. Avoid freezing.

KEEP OUT OF REACH OF CHILDREN. JAUHKAN DARIPADA KANAK-KANAK.

SHELF LIFE: Please refer to the outer package

PACK SIZE: Pack in 10 x 1 ml ampoules.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang,

Selangor, MALAYSIA.